Question	Answer	Marks
3(a)(i)	$a = (v^2 - u^2) / 2s$ $= (22^2 - 13^2) / (2 \times 180)$	C1
	$= 0.88\ \text{m s}^{-2}$	A1
	OR [$t = (180 \times 2) / (22 + 13) = 10.3$] $180 = 13 \times 10.3 + ½\ a \times 10.3^2$ or $180 = 22 \times 10.3 - ½\ a \times 10.3^2$ or $22 = 13 + a \times 10.3$	(C1)
	$a = 0.88\ \text{m s}^{-2}$	(A1)
3(a)(ii)	$(\Delta)E = ½m(\Delta)v^2$	C1
	gain in KE $= ½m(v^2 - u^2)$ $= ½ \times 9400 \times (22^2 - 13^2)$	C1
	$= 1.5 \times 10^6\ \text{J}$	A1
	OR $W = Fs$ $= ma \times d$	(C1)
	gain in KE $= 9400 \times 0.88 \times 180$	(C1)
	$= 1.5 \times 10^6\ \text{J}$	(A1)
3(b)(i)	rate of change of momentum	B1
3(b)(ii)	(Force =) $(2.5 \times 10^4 - 21 \times 10^4) / 15 = -1.2 \times 10^4$ (N)	A1

Question	Answer	Marks
3(b)(iii)	The change of momentum (for S and R to come to rest) is the same	**B1**
	(Average) force on R (to come to rest) is ($–21 \times 10^4 / 23$ =) 0.91×10^4 N **or** (Average) force on R (to come to rest) is less than the force on S / less than *F*	**B1**
	(S will come to rest in) less time (than R).	**B1**
	OR The change of momentum (for S and R to come to rest) is the same	**(B1)**
	Time for S to come to rest is ($–21 \times 10^4 / 1.2 \times 10^4$ =) 17.5 s (and time for R to come to rest is 23 s)	**(B1)**
	(S comes to rest in) less time (than R).	**(B1)**

Question	Answer	Marks